Differential Equations

Subject: Mathematics | NoteShare Library

DIFFERENTIAL EQUATIONS

1. FIRST ORDER
Separable: dy/dx = f(x)g(y)
Linear: dy/dx + P(x)y = Q(x)

2. SECOND ORDER
ay’ + by’ + cy = 0
Characteristic: ar + br + c = 0

Cases:

3. LAPLACE TRANSFORM

L{1} = 1/s
L{e^at} = 1/(s-a)
L{sin at} = a/(s +a)
L{cos at} = s/(s +a)

4. APPLICATIONS
- Population growth
- Spring-mass
- RC/RL circuits
- Heat equation